TL2GH^{2}T: Triple-Path Local-to-Global Network With Hybrid Head Transformer for Hyperspectral Change Detection

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***Abstract*— With the aid of transformers, significant progress has been achieved in hyperspectral image change detection (HSI-CD) in recent times. Nonetheless, most contemporary detection methods fail to incorporate diverse diagnostic features extracted from hyperspectral (HS) images. In addition, relying solely on algebraic-based techniques to extract information of difference is insufficient for achieving satisfactory detection performance. In this regard, we propose an innovative triple-path local-to-global network (TL2GN), complemented by a hybrid head transformer (HybridHT), called TL2GH^{2}T, tailored for HSI-CD tasks. To be specific, TL2GH^{2}T first investigates spatial, spectral, and spatial–spectral features from a local-to-global perspective. Then, a novel spatial and spectral token fusion (SSTF) module is developed to integrate the above three tokenized features, producing discriminative features from two HS images separately. Moreover, drawing inspiration from chromosomal crossover mechanisms, we propose a HybridHT. Its goal is to simultaneously learn cross correlation and self-correlation information of bitemporal features from a global perspective, producing highly discriminative distinctions. Our approach, validated through extensive experimentation on four varied HS benchmarks, exhibits exceptional performance in HSI-CD, outperforming contemporary methods in both visual and quantitative evaluations.**

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***Index Terms*— Change detection (CD), hybrid head transformer (HybridHT), hyperspectral (HS) image, transformer.**

I. Introduction

REMOTE sensing (RS) change detection (CD) offers valuable insights into the correlation between human production activities and surface changes [1], [2]. Imaging spectroscopy sensors yield abundant information by capturing a broad range of spectral bands, generating hyperspectral (HS) images that facilitate the characterization of subtle physical and chemical information in the Earth's surface [3], [4]. Consequently, HS image CD (HSI-CD) is used to assess ground changes by leveraging subtle variations in the reflectance spectra of land covers in bitemporal images, which is of great significance for refined detection and management of surface changes [5].

Acquired bitemporal HS images of the same region must undergo a series of preprocessing operations, such as geometric correction, radiometric correction, image alignment, and denoising, to mitigate the effects of pseudochanges induced by environmental factors before CD [6]. Subsequently, various CD algorithms can be applied to achieve the identification of change regions [7], serving as the research core focus of this article.

In the early stages of HSI-CD development, numerous algorithms using algebraic comparison, subspace learning, and object classification were extensively investigated. Regarding algebra-based methods, the change vector analysis (CVA) [8] stands out as the most representative approach, making change judgments based on the magnitude of the algebraic difference between the original spectral vectors of pixels at the same location. Although this method is friendly to apply, its detection performance is inevitably affected by the quality of the imaging and the setting of the change threshold. The subspace learning-based methods achieve difference recognition by transforming or mapping the original HS images into an easily distinguishable low-dimensional subspace [9], [10]. Specifically, commonly used detection algorithms based on subspace transformations include principal component analysis (PCA) [11], slow feature analysis (SFA) [12], multivariate

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alteration detection (MAD) [13], and iteratively reweighted MAD (IR-MAD) [14]. Moreover, detection methods based on classification techniques typically entail directly comparing two time-phase images and then using a classifier (e.g., extreme learning machine (ELM) [15], K-nearest neighbors (KNNs) [16], or support vector machine (SVM) [17]) to identify changes. Although the traditional methods mentioned above have achieved some success in the HSI-CD task, relying solely on handcrafted features makes it challenging to attain satisfactory detection performance, especially when dealing with complex detection scenarios.

In recent years, feature extraction paradigms based on deep learning (DL) have been extensively studied to acquire more discriminative features, achieving promising performances in various downstream tasks based on HS imagery, such as classification [18], [19], target detection [20], and spectral unmixing [21]. With a similar motivation, DL-based HSI-CD algorithms have also been widely explored. Various backbone networks, including convolutional neural networks (CNNs), graph neural networks (GNNs), recurrent neural networks (RNNs), and generative adversarial networks (GANs), have been introduced for HSI-CD tasks. As a potent instrument for abstract feature extraction, CNN-related technique swiftly evolves into the focal point of exploration in HSI-CD tasks. To be specific, we categorize CNN-related HSI-CD methods into the following categories: 1) algorithms combining CNNs with traditional methods; 2) algorithms based purely on CNNs; and 3) CNNs combined with other DL-based backbone algorithms.

By harnessing the powerful feature representation capabilities of convolutional techniques, traditional methods can achieve more promising performance. For instance, Saha et al. [22] constructed deep CVA (DCVA) using a pretrained CNN to provide more discriminative deep features for the CVA detector, yielding more accurate detection results. Similarly, Du et al. [23] used SFA to analyze changes in the depth spectral feature level, rather than the original spectral feature level, improving detection accuracy. Wu et al. [24] performed adaptive band selection by adopting a CNN, guided by CD results to prioritize highly discriminative bands. A more common strategy is to directly investigate the utilization of CNN models with different structures to achieve HSI-CD. For example, Wang et al. [25] proposed an HSI-CD framework based on end-to-end 2-D CNNs, which efficiently handles high-dimensional data by introducing a mixed affinity matrix, achieving outstanding results on real datasets. Lin et al. [26] achieved excellent detection results with customized bilinear CNNs (BCNNs) that efficiently learned feature representations of bitemporal images, capturing the correlation between the two inputs using a linear outer product operation. Zhan et al. [27] proposed a three-directional spectral–spatial CNN (TDSSC) for HSI-CD, which effectively extracts the spectral and spectral–spatial information by decomposing the change tensor in three directions. Similarly, Fang et al. [28] further proposed a spatial–spectral decoupled interaction network (SSDIN), aiming to extract spectral and spatial features separately in a manner of shared weights. Recently, attention mechanisms have been widely incorporated into HSI-CD algorithms, given their prominence in optimizing feature representative capabilities. For instance, Wang et al. [29] designed a convolutional block attention module (CBAM) in the feature extraction backbone, enabling adaptive extraction of spectral and spatial features while exploring bitemporal correlated features through the Siamese strategy. Moreover, Liu et al. [30] stacked multiple CBAM to construct a stacked attention module to generate more distinguishable feature pairs. Furthermore, Gong et al. [31] formulated spatial attention weights in Gaussian space to suppress noise and enhance relevant features in the difference maps of two HS images. With similar motivation, Yang et al. [32] attempted to use attentional mechanisms to enhance characteristics that are change-sensitive, yielding higher detection performance. To achieve clear detection in boundaries, Wang et al. [33] proposed a trainable superpixel algorithm that adaptively collaborates with DL-based networks for CD tasks.

To achieve more accurate detection results, additional DL backbones, including GCN, RNN, and GAN, were introduced to accompany convolutional architectures [34]. For instance, Wang et al. [35] proposed a novel graph attention model by integrating spatial location information with graph attention, enabling autonomous learning of spatial relationships among objects, and achieving robust detection results. Mou et al. [36] cascaded CNNs and RNNs to construct the recurrent CNN (ReCNN) to efficiently learn joint spectral–spatial–temporal features, yielding promising detection results. In addition, variants of RNNs, specifically the long short-term memory (LSTM) and gated recurrent unit (GRU), were also investigated to integrate with convolutional structures to boost the extraction of temporal correlations [37], [38]. Moreover, Wu et al. [39] developed a GAN-based detector using a fully convolutional network to achieve flexible detection under various supervised conditions.

Although promising detection performance has been achieved by these CNN-based backbone networks and their variants, the limited receptive field of the convolutional operation constrains a more comprehensive understanding of the input image. More recently, the transformer architecture, initially proposed in the field of natural language processing (NLP), has garnered attention in the field of computer vision (CV) for its efficient global feature extraction, giving rise to the groundbreaking vision transformer (ViT) [40]. Following this, a plethora of ViT-based approaches have been investigated in the field of CV and applied to various downstream tasks such as classification [41], [42], object recognition [43], and target detection [44].

Motivated by these achievements, the exploration of transformer-based HSI-CD algorithms has also been carried out extensively. Furthermore, transformer-based HSI-CD methods can be broadly categorized into two groups: those using only the transformer encoder for feature extraction and those integrating the transformer encoder with other backbones. Specifically, Ding et al. [45] used token splicing results from bitemporal images as input to a transformer encoder, facilitating correlation extraction between bitemporal features. Moreover, Zhang et al. [46] used transformer encoders to hierarchically extract spatial–temporal and spectral–temporal

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features, achieving impressive results. Notably, to enhance the accuracy of capturing changes, Wang et al. [47] introduced spectral–spatial–temporal transformers (SST-Former), specifically designing cross-attention to construct the temporal transformer encoder. Furthermore, Xu et al. [48] parallelly devised stacked and parallel pyramid transformer encoders to extract richer temporal correlation information. To compensate for the transformer's limited consideration of local features, some methods used the feature maps obtained by the convolutional operation as the input to the transformer encoder. For example, Zhou et al. [49] used Ghost module-based convolution to provide tokens embedded with local semantic information for transformer encoders in a lightweight manner. More directly, Zhang et al. [50] adopted HybridSN proposed by Roy et al. [51] to generate tokens for the spectralwise transformer encoder, achieving high-performance detection results. Moreover, Dong et al. [52] innovatively investigated the correlation of the dual temporal phase abundance matrix by taking advantage of the proposed self-cross-hybrid attention, mitigating the interference of mixed pixels. More recently, Zhang et al. [53] integrated the network architecture search strategy with the transformer encoder to achieve a more flexible feature representation process, boosting detection accuracy.

It must be acknowledged that the aforementioned DL-based methods have acquired decent performance in feature extraction from HS images by capturing abstract features in spatial or spectral dimensions. To accomplish the CD task with HS images, one can naively adopt backbones from the CV region to model spatial features. However, the comprehensiveness of features extracted by the above backbones is not sufficient enough, especially when it comes to spectral features. In fact, spatial, spectral, and spectral–spatial features are all available diagnostic features. Especially, spectral-related information is particularly effective for detecting changes occurring in HS images. Moreover, even if it is provided with representative features of an HS image, a model may still be unable to precisely detect changes by straightforward comparison. Unlike the processing of individual HS images, the CD task not only requires the extraction of discriminative features from individual HS image but also demands better capture of discrepant features in the dual-temporal images. In summary, there are still some pressing issues in the extraction of diagnostic features and discriminative change information that need to be addressed. To be specific, the existing problems are summarized as follows.

1) Due to the unique structure of HS images, diagnostic features can manifest as spatial, spectral, and spatial–spectral features. However, prevailing CNN-transformer-based methodologies predominantly extract diagnostic features solely in the spatial–spectral domain, leading to incomplete representation. Moreover, the fusion of these three feature forms in a customized manner has rarely been studied.
2) Most current HSI-CD algorithms typically use algebraic-based methods (e.g., differencing, summing, or concatenating) to compare features from bitemporal images for change result evaluation, lacking the capability to learn highly representative confident in the global perspective.

To address the aforementioned issues, we propose a CNN-transformer-based network specifically designed for HSI-CD, named TL2GH^{2}T. Specifically, the proposed TL2GH^{2}T is able to acquire a comprehensive feature representation of HS images in spatial, spectral, and spatial–spectral dimensions, ensuring the accuracy of later difference information extraction. In addition, a hybrid head transformer (HybridHT) tailored for CD tasks is also proposed to extract temporal differences from the viewpoint of temporal global correlation.

More specifically, our contributions are summarized as follows.

1) We propose a novel CD method, TL2GH^{2}T, based on a hybrid CNN-transformer architecture specifically designed for HSI-CD. This method meticulously extracts diagnostic features from bitemporal HS images, enabling the acquisition of highly reliable difference information crucial for precise detection. Furthermore, we validate its state-of-the-art performance across four real-world scenarios.
2) We devise a simple but effective feature extraction network, referred to as the triple-path local-to-global network (TL2GN), with the goal of simultaneously learning features from spatial, spectral, and spatial–spectral dimensions from local-to-global perspectives of HS image. Moreover, we introduce a novel spatial and spectral token fusion (SSTF) module to obtain more comprehensive and diagnostic features for subsequent extraction of change information.
3) Motivated by the principles of chromosomal crossover, we propose a new transformer-based module, i.e., HybridHT, which can hierarchically produce high discriminative difference information. To be specific, the HybridHT encoder (H^{2}TE) models cross correlation and self-correlation of bitemporal features globally at each stage, enhancing discriminative feature extraction for subsequent stages.

The remainder of this article is organized as follows. Section II provides a brief preliminary of the multiscale self-attention (SA). Section III presents a detailed description of the proposed TL2GH^{2}T framework. Section IV conducts extensive experiments on four datasets, including ablation studies and comparative analyses with the existing state-of-the-art methods. Finally, Section V gives a conclusion and highlights the main contributions of our work.

II. Preliminaries of Multihead SA

A. SA Mechanism

The attention mechanism used in DL-based models is inspired by human visual processing, allowing it to focus more on the most valuable parts of the visual field. Likewise, models that use the attention mechanism can assign distinct weights to various segments of the input data, thereby aiding in the extraction of more discriminative features to inform the final decision. In contrast to the attention mechanism, the SA mechanism explores the correlation of each input element (e.g., pixel or word), as depicted in the right of Fig. 1.

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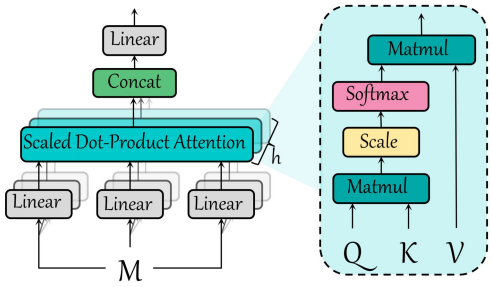


Fig. 1. Diagram of MHSA.

For an input sequential feature map $\mathbf{M} = [\mathbf{m}_1, \mathbf{m}_2, \ldots, \mathbf{m}_n]$, the overall SA procedure can be expressed as follows:

$$\mathbf{Q} = \mathbf{M} \times \boldsymbol{\omega}_q, \mathbf{K} = \mathbf{M} \times \boldsymbol{\omega}_k, \mathbf{V} = \mathbf{M} \times \boldsymbol{\omega}_v \quad (1)$$

$$\mathrm{SA}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \mathrm{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d}}\right)\mathbf{V} \quad (2)$$

where $\mathbf{M}$ is initially mapped with three different projection matrices $\boldsymbol{\omega}_q$, $\boldsymbol{\omega}_k$, and $\boldsymbol{\omega}_v$, respectively, to provide the query ($\mathbf{Q} = [\mathbf{q}_1, \mathbf{q}_2, \ldots, \mathbf{q}_n]$), key ($\mathbf{K} = [\mathbf{k}_1, \mathbf{k}_2, \ldots, \mathbf{k}_n]$), and value ($\mathbf{V} = [\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n]$) sets. Then, the attention weight score of SA is obtained by calculating the inner product of $\mathbf{Q}$ and $\mathbf{K}$, followed by the softmax activation function. To prevent gradient vanishing, it is noteworthy that a normalization process is commonly applied on $\mathbf{Q}\mathbf{K}^T$, i.e., $(\mathbf{Q}\mathbf{K}^T/(d)^{1/2})$, where d represents the dimension of $\mathbf{q}_i$ or $\mathbf{k}_i$. Finally, the reweighted representation for the input feature is obtained.

B. Multihead SA

As illustrated in Fig. 1 (left), to enrich feature representation, multiple aforementioned SA modules are adopted in parallel (so-called multihead). Mathematically, the multihead SA ((MHSA) can be simply formulated as

$$\mathrm{MHSA}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \mathrm{Concat}(\mathrm{head}_1, \ldots, \mathrm{head}_h) \times \boldsymbol{\omega}_{\mathrm{o}} \quad (3)$$

where $\mathrm{head}_i = \mathrm{SA}(\mathbf{Q}, \mathbf{K}, \mathbf{V})$, h and $\boldsymbol{\omega}_{\mathrm{o}}$ denote the number of SA and linear projection function, respectively. Concat(,) is the concatenate function. This allows for flexible attention to the features from various subspaces.

III. Methodology

In this section, we first provide a brief overview of its overall architecture. Then, we elaborate the details of the essential components of the proposed approach, which includes local feature embedding for transformer encoders, triple-parallel transformers designed for spectral, spatial, and spectral–spatial long-range features extraction, and finally, a customized HybridHT for temporal change features' exploration.

A. Overview of the Proposed TL2GH^{2}T

As depicted in Fig. 2, the proposed HSI-CD framework is designed in a dual-stage manner to better capture the temporal spectral–spatial features between two HS images taken at two different times (i.e., T_1 and T_2). In simpler terms, our proposed model uses two of the identical subnetworks (i.e., TL2GN) to first obtain high-level representative features from two images and then HybridHT calculates the difference between these two representations. It can be observed that 3-D patches generated from two HS images are used as the network training samples, which ensures the utilization of both spectral and spatial information. For an input patch of the proposed TL2GH^{2}T, their three different local feature embeddings (i.e., spatial, spectral, and spatial–spectral features) are initially generated by three parallel convolutional operations. Then, the obtained feature embeddings are tokenized and transmitted into different transformer encoders to explore long-range correlations within the aforementioned feature embeddings. After that, a simple and generic triple-path fusion module (i.e., SSTF) is designed to produce highly discriminative and comprehensive spectral–spatial features. In the subsequent stage, the proposed HybridHT, specifically designed for the CD task, effectively extracts the temporal differences between the two HS images by modeling long-range differential relationships seen in the features extracted in the previous stage. It is worth mentioning that the operations on the two temporal images before the final stage are identical. To keep things simple, we do not make a specific distinction between these operations.

B. Triple-Path Local-to-Global Network

1) Local Feature Extraction: As mentioned above, the input patches for our proposed model are cropped from the two dimension-reduced HS images $\mathbf{X}_{T_{1/2}} \in \mathbb{R}^{H\times W\times C}$, where $H \times W$ and C denote the spatial height and width and the amount of spectral information, respectively. Notably, the original HS image contains B spectral bands, which were reduced to C principle components using the widely used PCA, ensuring low computer complexity. Specifically, the obtained patches are extracted with its $p \times p$ spatial neighborhood pixels, which can be denoted as $\mathbf{x}_i^{T_{1/2}} \in \mathbb{R}^{p\times p\times C}$, $i \in (1, \ldots, H \times W)$. In this way, the spectral and spatial features that existed in these neighborhood cubes can be investigated in the process that follows.

Specifically, three 3-D convolutional operations with different structuring elements are applied to explore the local spectral, spatial, and spectral–spatial features for each cube in parallel. Mathematically, a 3-D convolutional operation with a kernel of size (h, w, d) can be expressed as follows:

$$\mathbf{v}_{i,j}^{x,y,z} = \phi\left(\sum_{c}\sum_{\alpha=0}^{h_i-1}\sum_{\beta=0}^{w_i-1}\sum_{\gamma=0}^{d_i-1} k_{i,j,c}^{\alpha',\beta',\gamma'} \times \mathbf{v}_{(i-1),c}^{(x+\alpha')(y+\beta')(z+\gamma')} + b_{i,j}\right) \quad (4)$$

where $\mathbf{v}_{i,j}^{x,y,z}$ represents the calculated value located at (x, y, z) of the jth input feature cube calculated at the ith layer, and $k_{i,j}^{\alpha',\beta',\gamma'}$ and $b_{i,j}$ denote the weight parameter at position $(\alpha', \beta', \gamma')$ of the input feature cube and the bias. The c refers the indexes of the input feature cubes, and $\phi(\cdot)$ denotes the activation function, which is set to rectified linear unit (ReLU) throughout this article. Furthermore, the batch normalization (BN) is applied before the activation function to accelerate the convergence. The (h, w, d) of three 3-D convolution layers in

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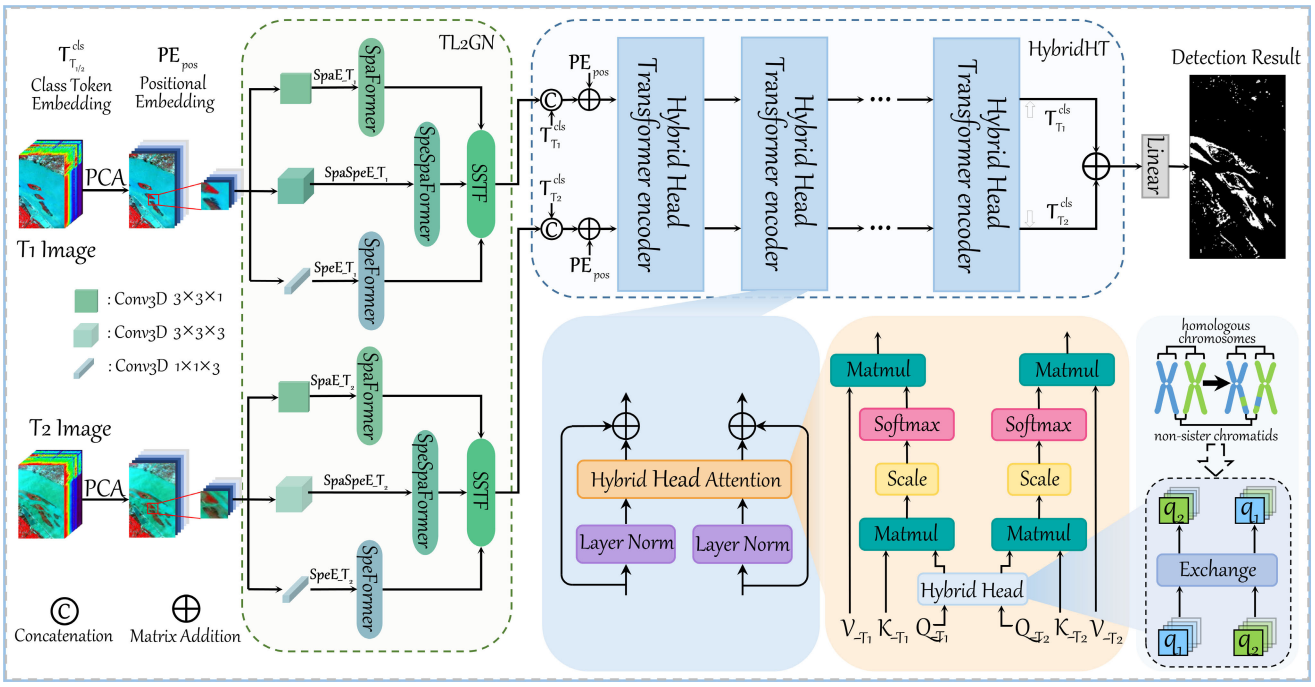


Fig. 2. Overview of the proposed TL2GH²T, which takes patches from two temporal as the input to learn their difference. The TL2GN is proposed to extract the diagnostic spatial–spectral features. The HybridHT is proposed to produce high discriminative difference information hierarchically by learning cross correlation and self-correlation of bitemporal features globally.

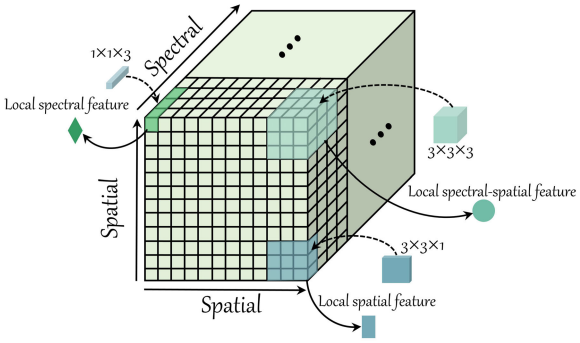


Fig. 3. Schematic of local spatial, spectral, and spatial–spectral feature extraction.

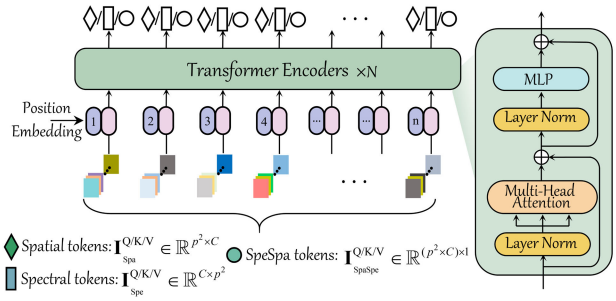


Fig. 4. Diagram of SpaFormer, SpeFormer, and SpeSpaFormer.

each subnetwork of this model are set to (1, 1, 3), (3, 3, 1), and (3, 3, 3), respectively. In addition, the same padding is applied and the stride in convolutional operation is set to (1, 1, 1). As seen in Fig. 3, three aspects of local features in the HS image are embedded using these three distinct 3-D convolutions.

The main reasons for selecting the aforementioned three different convolution operations as the initial process for the HS image can be summarized in three respects. First, the information provided by HS images is varied in different dimensions, which motivates us to comprehensively investigate them to obtain diagnostic information for the final decision. Furthermore, the transformers are used afterward, primarily concerned with global signature relationships, while ignoring the local texture feature. Finally, the 3-D properties of HS images render them inappropriate for direct input to the transformer after a simple linear mapping technique, as this could negatively impact the features' discriminability.

2) Global Feature Extraction: As depicted in Fig. 2, after the local semantic features from the three perspectives are extracted (i.e., **SpaE_T$_{1/2}$**, **SpeE_T$_{1/2}$**, and **SpeSpaE_T$_{1/2}$**), the global correlations under these perspectives are subsequently explored. Similar to the local feature extraction phase, the global feature extraction phase consists of three parallel modules designed to explore highly discriminative long-range features of each image from three aspects simultaneously, facilitating subsequent temporal information extraction.

As shown in Fig. 4, the spectral transformer (SpeFormer), spatial transformer (SpaFormer), and spectral–spatial transformer (SpeSpaFormer) are introduced to learn spectral and spatial global features by directly adopting the MHSA module. Then, we rearrange the obtained spatial feature embedding, i.e., **SpaE_T$_{1/2}$** $\in \mathbb{R}^{p\times p\times C}$, into pixel token sequence $\mathbf{x}_{\text{SpaE}}^{T_{1/2}} = \{\mathbf{x}_{\text{SpaE}}^{1}, \mathbf{x}_{\text{SpaE}}^{2}, \ldots, \mathbf{x}_{\text{SpaE}}^{p^2}\}$, where $\mathbf{x}_{\text{SpaE}}^{T_{1/2}} \in \mathbb{R}^{p^2\times C}$. Afterward, using $\mathbf{X}_{T_1}$ as an example image, the **Q**/**K**/**V** matrices, i.e., $\mathbf{I}_{\text{spa}}^{Q/K/V}$ of the MHSA module are obtained by first adding an addition operation with position encoding and followed by applying linear projection functions on $\mathbf{x}_{\text{SpaE}}^{T_1}$. Mathematically,

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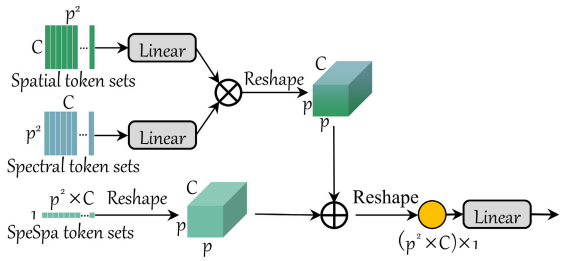


Fig. 5. Diagram of SSTF module.

these can be formulated as follows:

$$\mathbf{I}_{\text{spa}}^{Q} = \left(\mathbf{x}_{\text{SpaE}}^{T_1} + \mathbf{PE}_{\text{spa}}\right) \times \boldsymbol{\omega}_{\text{spa}}^{q} \tag{5}$$

$$\mathbf{I}_{\text{spa}}^{K} = \left(\mathbf{x}_{\text{SpaE}}^{T_1} + \mathbf{PE}_{\text{spa}}\right) \times \boldsymbol{\omega}_{\text{spa}}^{k} \tag{6}$$

$$\mathbf{I}_{\text{spa}}^{V} = \left(\mathbf{x}_{\text{SpaE}}^{T_1} + \mathbf{PE}_{\text{spa}}\right) \times \boldsymbol{\omega}_{\text{spa}}^{v} \tag{7}$$

where $\boldsymbol{\omega}_{\text{spa}}^{\text{q/k/v}} \in \mathbb{R}^{C\times C}$ is the trainable weight matrix of linear projection function, and $\mathbf{PE}_{\text{spa}} \in \mathbb{R}^{p^2\times C}$ denotes the position encoding matrix. Then, on $\mathbf{I}_{\text{spa}}^{\text{Q/K/V}}$, the layer normalization (LN) operation, MHSA module, and multilayer perceptron (MLP) module are applied to obtain the final output $\mathbf{Z}_{\text{spa}}$ of SpaFormer encoder, which can be expressed as follows:

$$\mathbf{Z}'_{\text{Spa_}T_1} = \text{MHSA}\left(\text{LN}\left(\mathbf{I}_{\text{spa}}^{Q}, \mathbf{I}_{\text{spa}}^{K}, \mathbf{I}_{\text{spa}}^{V}\right)\right) + \left(\mathbf{I}_{\text{spa}}^{Q}, \mathbf{I}_{\text{spa}}^{K}, \mathbf{I}_{\text{spa}}^{V}\right) \tag{8}$$

$$\mathbf{Z}_{\text{Spa_}T_1} = \text{MLP}\left(\text{LN}\left(\mathbf{Z}'_{\text{spa_}T_1}\right)\right) + \mathbf{Z}'_{\text{spa_}T_1} \tag{9}$$

where the residual connection is also used to alleviate gradient vanishing. As a result, the spatial semantic feature (i.e., $\mathbf{Z}_{\text{Spa_}T_{1/2}} \in \mathbb{R}^{p^2\times C}$) is obtained with a global receive field.

Similarly, we rearrange the obtained spectral feature embedding $\mathbf{SpeE_T_{1/2}} \in \mathbb{R}^{p\times p\times C}$, into spectral token sequence $\mathbf{x}_{\text{SpeE}}^{T_{1/2}} = \{\mathbf{x}_{\text{SpeE}}^{1}, \mathbf{x}_{\text{SpeE}}^{2}, \ldots, \mathbf{x}_{\text{SpeE}}^{C}\}$, where $\mathbf{x}_{\text{SpeE}}^{T_{1/2}} \in \mathbb{R}^{C\times p^2}$. For the spectral–spatial feature embedding $\mathbf{SpeSpaE_T}_{1/2} \in \mathbb{R}^{p\times p\times C}$, we rearrange them into pixel token sequence $\mathbf{x}_{\text{SpeSpaE}}^{T_{1/2}} = \{\mathbf{x}_{\text{SpeSpaE}}^{1}, \mathbf{x}_{\text{SpeSpaE}}^{2}, \ldots, \mathbf{x}_{\text{SpeSpaE}}^{p^2\times C}\}$, where $\mathbf{x}_{\text{SpeSpaE}}^{T_{1/2}} \in \mathbb{R}^{(p^2\times C)\times 1}$. Then, SpeFormer and SpeSpaFomer are constructed by the aforementioned similar architecture of SpaFormer and which operated on input $\mathbf{x}_{\text{SpeE}}^{T_{1/2}}$ and $\mathbf{x}_{\text{SpeSpaE}}^{T_{1/2}}$, to generate the global spectral and spectral–spatial semantic feature (i.e., $\mathbf{Z}_{\text{Spe_}T_{1/2}} \in \mathbb{R}^{C\times p^2}$ and $\mathbf{Z}_{\text{SpeSpe_}T_{1/2}} \in \mathbb{R}^{(p^2\times C)\times 1}$), respectively.

3) Spatial and Spectral Token Fusion: After extracting three long-range weighted correlations of spatial, spectral, and spectral–spatial as previously discussed, we further consider the fusion of those to generate comprehensive features of a single temporal image. As a result, we introduce a token fusion module, i.e., SSTF, which is specifically designed to integrate information from different-sized token sets (i.e., $\mathbf{Z}_{\text{Spa_}T_{1/2}}$, $\mathbf{Z}_{\text{Spe_}T_{1/2}}$, and $\mathbf{Z}_{\text{SpeSpe_}T_{1/2}}$). As depicted in Fig. 5, two linear functions with the weight of $\boldsymbol{\omega}_{\text{spa_}T_{1/2}} \in \mathbb{R}^{C\times 1}$ and $\boldsymbol{\omega}_{\text{spe_}T_{1/2}} \in \mathbb{R}^{p^2\times 1}$ are primarily adopted to project the spatial and spectral token set into $\widehat{\mathbf{Z}}_{\text{Spa_}T_{1/2}} \in \mathbb{R}^{p^2\times 1}$ and $\widehat{\mathbf{Z}}_{\text{Spe_}T_{1/2}} \in \mathbb{R}^{C\times 1}$. Then, the deep correlation among the spectral and spatial features can be further modeled as

$$\mathbf{A}_{\text{SpaSpe_}T_{1/2}} = \widehat{\mathbf{Z}}_{\text{Spa_}T_{1/2}} \otimes \left(\widehat{\mathbf{Z}}_{\text{Spe_}T_{1/2}}\right)^{T} \tag{10}$$

where $\mathbf{A}_{\text{SpaSpe_}T_{1/2}} \in \mathbb{R}^{p^2\times C}$ and $\otimes$ means inner product operation. After that, we simultaneously rearrange $\mathbf{A}_{\text{SpaSpe_}T_{1/2}}$ and $\mathbf{Z}_{\text{SpaSpe_}T_{1/2}}$ into $\widehat{\mathbf{A}}_{\text{SpaSpe_}T_{1/2}} \in \mathbb{R}^{p\times p\times C}$ and $\widehat{\mathbf{Z}}_{\text{SpaSpe_}T_{1/2}} \in \mathbb{R}^{p\times p\times C}$ and add them together to produce a highly discriminative spectral–spatial representation $\widehat{\mathbf{Z}}_{T_{1/2}} \in \mathbb{R}^{p\times p\times C}$ for the final differential comparison. Furthermore, we rearrange $\widehat{\mathbf{Z}}_{T_{1/2}}$ into the size of $\dot{Z}_{T_{1/2}} \in \mathbb{R}^{(p^2\times C)\times 1}$ and then project it to $\mathbf{Z}_{T_{1/2}}$ having the size of $(p^2\times C)\times d$, yielding spectral–spatial tokens for the next stage input, which can be formulated as

$$\mathbf{Z}_{T_{1/2}} = \dot{Z}_{T_{1/2}} \times \boldsymbol{\omega}_{T_{1/2}} \tag{11}$$

where $\boldsymbol{\omega}_{T_{1/2}} \in \mathbb{R}^{1\times d}$ is the weight of linear projection function, and d is the dimension of the latent feature for each token.

C. Temporal Correlation Extraction

The capability of accurately extracting features that distinguish between two images is also essential for CD, and this heavily depends on the discriminative power of prior feature extraction and the technique used for difference extraction. Traditionally, difference information is obtained through direct feature comparison, which is quick and easy to use but often fails to capture subtle changes in an efficient manner. To explore possible differences, we introduce the HybridHT, an innovative strategy to capture the difference between the extracted feature inside TL2GH^{2}T.

As a result, we create a transformer-like structure with several newly constructed H^{2}TEs to deal with the serialized features that were obtained in the preceding stage (Fig. 2). Similar to the operation mentioned in Section III-B, we first tokenize $\mathbf{Z}_{T_1}$ and $\mathbf{Z}_{T_2}$ for the subsequent attention modules, which can be formulated as follows:

$$\mathbf{z}_{T_1} = \left[\mathbf{T}_{T_1}^{\text{cls}},\ \mathbf{Z}_{T_1}\right] + \mathbf{PE}_{\text{pos}} \tag{12}$$

$$\mathbf{z}_{T_2} = \left[\mathbf{T}_{T_2}^{\text{cls}},\ \mathbf{Z}_{T_2}\right] + \mathbf{PE}_{\text{pos}} \tag{13}$$

where $\mathbf{PE}_{\text{pos}} \in \mathbb{R}^{(p^2\times C+1)\times d}$ is the position embedding, and $\mathbf{z}_{T_1} \in \mathbb{R}^{(p^2\times C+1)\times d}$ and $\mathbf{z}_{T_2} \in \mathbb{R}^{(p^2\times C+1)\times d}$ represent obtained sequential token sets for the followed hybrid head attention (HHA) block. Unlike the previous SpaFormer, we concatenate additional class token $\mathbf{T}_{T_{1/2}}^{\text{cls}} \in \mathbb{R}^{1\times d}$ in the respective token sequence, which is the extra trainable token to learn the difference produced from H^{2}TE. Furthermore, the input $\mathbf{Q/K/V}$ matrices of the HHA is obtained by applying linear projection functions $\boldsymbol{\omega}_{T_{1/2}}^{\text{Q/K/V}} \in \mathbb{R}^{d\times d}$ on $\mathbf{z}_{T_{1/2}}$, which can be mathematically expressed as

$$\mathbf{Q}_{T_{1/2}} = \mathbf{z}_{T_{1/2}} \times \boldsymbol{\omega}_{T_{1/2}}^{Q} \tag{14}$$

$$\mathbf{K}_{T_{1/2}} = \mathbf{z}_{T_{1/2}} \times \boldsymbol{\omega}_{T_{1/2}}^{K} \tag{15}$$

$$\mathbf{V}_{T_{1/2}} = \mathbf{z}_{T_{1/2}} \times \boldsymbol{\omega}_{T_{1/2}}^{V}. \tag{16}$$

Subsequently, drawing insights from the chromosomal crossover process, we propose the HHA, which can explore the correlation of two inputs from a global perspective. Chromosomal crossover refers to the exchange of genetic information

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between nonsister chromatids of homologous chromosomes during sexual reproduction. Consequently, this process results in the formation of recombinant chromosomes, significantly enhancing genetic diversity. Similarly, we conceptualize two parallel MHSA modules as homologous chromosomal nonsister chromatid monomers for the purpose of differential exploration where the genetic material is represented by the multiple heads they contain. Following that, a procedure similar to chromosome crossover interchange is carried out by exchanging query sets in the partial heads. The exchange of partial query sets aims to acquire more discriminative differential features, facilitating the exploration of correlations between distinct temporal images. More specifically, the detailed process through the HHA can be mathematically expressed as follows:

$$\boldsymbol{\alpha}^{m}_{T_2T_1} = \mathrm{SA}\big(\mathbf{Q}_{T_2}, \mathbf{K}_{T_1}, \mathbf{V}_{T_1}\big), \boldsymbol{\alpha}^{n}_{\mathbf{T_1}T_1} = \mathrm{SA}\big(\mathbf{Q}_{T_1}, \mathbf{K}_{T_1}, \mathbf{V}_{T_1}\big) \tag{17}$$

$$\boldsymbol{\alpha}^{m}_{T_1T_2} = \mathrm{SA}\big(\mathbf{Q}_{T_1}, \mathbf{K}_{T_2}, \mathbf{V}_{T_2}\big), \boldsymbol{\alpha}^{n}_{\mathbf{T_2}T_2} = \mathrm{SA}\big(\mathbf{Q}_{T_2}, \mathbf{K}_{T_2}, \mathbf{V}_{T_2}\big) \tag{18}$$

$$\mathbf{O}_{T_1} = \mathrm{Concat}\big(\boldsymbol{\alpha}^{m}_{T_2T_1},\ \boldsymbol{\alpha}^{n}_{T_1T_1}\big) \times \boldsymbol{\omega}^{\mathrm{O}}_{T_1} \tag{19}$$

$$\mathbf{O}_{T_2} = \mathrm{Concat}\big(\boldsymbol{\alpha}^{m}_{T_1T_2},\ \boldsymbol{\alpha}^{n}_{T_2T_2}\big) \times \boldsymbol{\omega}^{\mathrm{O}}_{T_2} \tag{20}$$

where $m \in [1, \ldots, \delta]$ and $n \in [\delta+1, \ldots, h]$ are the number of the head, respectively. The δ represents the degree of hybrid, signifying the extent of exploration of interrelated information between two temporal images in each multihead attention module, and $\boldsymbol{\omega}^{\mathrm{O}}_{T_{1/2}} \in \mathbb{R}^{(h\times d)\times d}$ is the linear projection function. Therefore, H²TE can be formulated as

$$\mathrm{H^2TE}\big(\mathbf{z}_{T_1}, \mathbf{z}_{T_2}\big) = \mathrm{HHA}\big(\mathrm{LN}\big(\mathbf{z}_{T_1}\big),\ \mathrm{LN}\big(\mathbf{z}_{T_2}\big)\big) + \big(\mathbf{z}_{T_1}, \mathbf{z}_{T_2}\big). \tag{21}$$

Notably, in contrast to the standard MHSA, the SA weights within each HHA module are not determined only by the query and key sets of individual images. As suggested in the above process, to determine the global contextual weights, we not only considered the global SA but also thoroughly investigated the interrelation between the two temporal images. Furthermore, our proposed HHA not only extracts differential information but also preserves the original information from individual temporal phases, unlike SSTFomer. In addition to preventing the loss of original data, the technique of introducing mixed heads gradually gains more abstract single-temporal features enabling more accurate change information extraction in later stages. After passing through several H²TEs, furthermore, we first add the desired concatenated class token $\mathbf{T}^{\mathrm{cls}}_{T_1}$ and $\mathbf{T}^{\mathrm{cls}}_{T_2}$, which aid in learning discriminative temporal information, for the final change prediction. Then, we next pass this through a fully connected layer followed by a sigmoid function to obtain the final predicted decision $\mathbf{F}_{\mathrm{CD}}$, which can be formulated as

$$\mathbf{F}_{\mathrm{CD}} = \mathrm{FC}\big(\mathbf{T}^{\mathrm{cls}}_{T_1} + \mathbf{T}^{\mathrm{cls}}_{T_2}\big) \tag{22}$$

where FC(·) denotes a linear layer preceded by a BN and a ReLU layer.

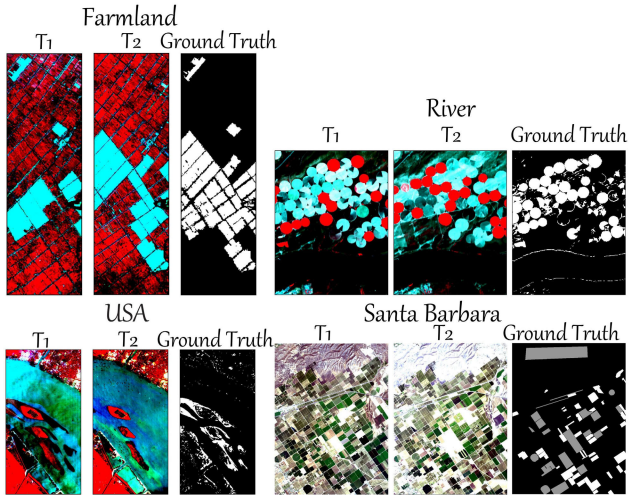


Fig. 6. Pseudocolor images and corresponding regions of change for the four scenarios.

IV. Experiments

This section conducts an extensive comparison with several other state-of-the-art algorithms on four generally recommended datasets to assess the detection performance of the proposed TL2GH²T. Furthermore, to facilitate the understanding of our experimental procedure, we will first present the details of the datasets, comparison methods, and experimental setup hierarchically. Then, we further discuss our proposed TL2GH²T in terms of parameters and modules. Finally, a comprehensive comparison with other state-of-the-art methods is implemented.

A. Dataset Description

In the following comprehensive experiments, four datasets, i.e., Farmland, River, USA, and Santa Barbara, are adopted. Fig. 6 displays pseudocolor images for these scenarios at two distinct time spans, along with the corresponding ground-truth map delineating the changed area.

1) The bitemporal images of the Farmland dataset consist of 450 × 140 pixels and nearly 155 continuous spectral bands. Its ground sampling distance (GSD) is about 30 m.
2) The River dataset is made up of 463 × 241 pixels along with 198 spectral bands. Its wavelength range is 0.4–2.5 μm with a GSD of 30 m.
3) The USA scene includes 16 070 changed pixels and 57 917 unchanged pixels within the spatial range of 307 × 241 pixels. To facilitate downstream tasks, this dataset reserves 154 spectral bands by removing several low signal-to-ratio bands.
4) Two images of the Santa Barbara dataset comprise 224 different spectral bands with 984 × 740 pixels and GSD of 20 m. Furthermore, this scene consists of 52 134 changed pixels versus 80 418 unchanged pixels.

Moreover, there are two categories (i.e., changed and unchanged) of land covers that need to be identified with the CD task. Therefore, a random selection of training and test samples is used, with the training ratio set to 2% for each of

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TABLE I
NUMBERS OF SAMPLES IN TRAINING AND TESTING SETS FOR THE FOUR DATASETS

Data set	Training Set			Test Set		
	Unchanged	Changed	Total	Unchanged	Changed	Total
Farmland	983	277	1260	48140	13600	61740
River	2063	169	2232	101072	8279	109351
USA	1159	321	1480	56758	15749	72507
SantaBarbara	1608	1042	2650	78810	51092	129902

two classes for all datasets, as reported in Table I to analyze the effectiveness of the proposed method.

B. Experimental Setup

1) Evaluation Metrics: In this study, both quantitative and qualitative evaluation criteria are adopted for evaluating the detection performance of the proposed method. In terms of quantitative assessment, overall accuracy (OA) and *kappa coefficient* generated by the confusion matrix are used to jointly evaluate the detection ability. On the other hand, we also provide the visualization result depicting the spatial distribution of the changed areas for a qualitative evaluation.

2) Comparison With State-of-the-Art Methods: To obtain a comprehensive and reliable evaluation of our proposed methodology, eight representative methods were meticulously selected for comparison, including CVA [8], PCA-CVA [11], recurrent 3-D fully convolutional networks (Re3FCNs) [54], SSDIN [28], GETNET [25], TDSSC [27], SST-Former [47], and CDFormer [45], respectively. The traditional machine learning models[1] for HSI-CD are available and open-source.

3) Implementation Details: This study uses an RTX 3090 GPU to accelerate all the experiments, which are run on the PyTorch 1.10.0 platform. Moreover, the Adam [55] optimizer is chosen to optimize the parameters of our proposed model, with the initial learning rate set to $1e^{-4}$. In addition, the learning rate was updated by multiplying 0.9 after one-tenth of the total epochs. In our experiments, the proposed model was trained for 100 epochs on all the datasets with the mini-batch size set to 64 to fit all the datasets properly. Furthermore, the number of heads of SpeFormer, SpaFormer, SpeSpaFormer, and HybridHT is empirically set to eight.

C. Discussion of Parameters

As discussed before, the size of the cropped 3-D patch (i.e., p and C), which determines the information fed into the proposed model, might affect the detection performance. As a result, we carried out extensive experiments over four datasets to search for the optimal setting of these two hyperparameters for further discussion. In terms of the model's structure-related hyperparameters, we evaluate the fluctuations in detection performance under the influence of depth of Spe-, Spa-, SpeSpaFormer, and HybridHT. Moreover, the hyperparameter δ governs the degree of interaction between two temporal features, dictating the extent to which correlation information is extracted. Therefore, we further investigate the affection of the value of this hyperparameter on detection performance.

[1]https://github.com/ChenHongruixuan/ChangeDetectionRepository

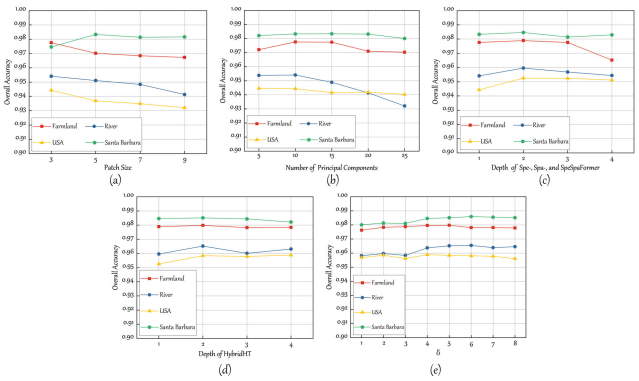


Fig. 7. Analysis (regarding OA) of TL2GH^{2}T with different hyperparameters on the four datasets. (a) Effect of different patch sizes. (b) Effect of different numbers of principal components. (c) Effect of different depths of Spe-, Spa-, and SpeSpaFormer. (d) Effect of different depths of HybridHT. (e) Effect of different δ.

1) Effect of Input Patch Size: As depicted in Fig. 7, we investigated the effect of the variable p on detection performance of four datasets in terms of OAs when p is in the range of $[3, 5, \ldots, 9]$. Notably, C and δ were set to 10 and 5, respectively, when p was analyzed. Overall, there is a similar trend in the impact of p on detection performance across the four datasets. To be more precise, with the exception of the Santa Barbara dataset, the detection accuracy decreases as p is increased. For the Farmland, River, and USA datasets, the gradual decrease in detection accuracy is attributed to their poor spatial resolution. On the other hand, the detection accuracy on the Santa Barbara dataset remains stable after the increase due to its higher spatial resolution. Following the analysis of experimental results, we have concluded that the optimal accuracy is obtained when p is set to 3 for the Farmland, River, and USA datasets, and 5 for the Santa Barbara dataset, respectively.

2) Effect of the Number of the Reserved Principal Components: Expanding on the previously mentioned analysis, we carried out an additional study to investigate the impact of principal components on detection performance. As depicted in Fig. 7, while the detection accuracy of our proposed model exhibits low sensitivity to variations in C, discernible regulars emerge in the influence trends associated with C. In particular, using fewer principal components leads to suboptimal performance across all the datasets, attributed to the significant loss of information. Furthermore, with the incorporation of an increased number of principal component features, a notable enhancement in detection performance is observed. However, it is imperative to highlight that better model performance does not always positively correlate with increasing the number of principal components. This is attributed to the fact that an abundance of principal components can encapsulate excessive and irrelevant information, thereby compromising model performance. Ultimately, we fixed C at 10 across all the datasets' applications.

3) Effect of Depth of Spe-, Spa-, and SpeSpaFormer: We evaluate the influence of the depth of these three feature extraction modules on detection performance by varying the number of transformer encoders. To be more precise, Fig. 7 illustrates the corresponding detection performance of the

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model on the four datasets. The depth (number of encoders) is set sequentially from 1 to 4. It can be observed that the number of encoders has a clear and similar impact on detection performance across the four datasets. Hence, we consistently set the number of encoders for these three feature extraction modules to two based on the detection performance evaluated across the four datasets.

4) Effect of depth of HybridHT: Similar to Section IV-C4, we further delved into the impact of using varying numbers of H^{2}TE on detection performance of the proposed model. The detection results (in terms of OA) of the model under different numbers of H^{2}TE are depicted in Fig. 7. Notably, the sensitivity of the model's detection accuracy to the changes in depth varies across different datasets. In particular, using two H^{2}TEs instead of one yields a more significant performance gain for the USA and River datasets. On the other hand, the performance of the model was less affected by the differences in depth for the other two datasets. However, with further increases in depth, the detection accuracy of the proposed model stabilizes across all the four datasets. In light of the aforementioned analysis and its practical implications, we used two H^{2}TEs in the proposed TL2GH^{2}T for our experiments across all the datasets.

5) Effect of the Value of δ: This hyperparameter plays a pivotal role in determining the extent to which intertemporal change correlation acquisition further determines the detection accuracy of the proposed model. As a result, we investigate this hyperparameter's sensitivity in parallel for each of the four datasets. Fig. 7 depicts the trend in variations in detection performance influenced by the value of δ in the range of $[1, 2, \ldots, 8]$. Overall, fluctuations in δ did not have a significant impact on detection performance. However, carefully observing the patterns in Fig. 7, we discern analogous trends across all four datasets. To be more precise, the performance of the proposed model exhibits some improvement across all the four datasets as δ increases. This is because exchanging the query set heads provides the potential to explore temporal change information from a long-range perspective. However, this upward trend does not persist when sufficient discrepancy information has already been captured, i.e., the value of δ reaches some extent. It can be summed up as follows: detection performance will be hampered if intertemporal change information is overemphasized at the expense of self-correlation that exists within each temporal feature separately. Therefore, combining the description in Fig. 7 and the aforementioned analysis, we uniformly set δ to 5 across all the four datasets.

D. Ablation Discussion

We further investigate the effect of the customized HybridHT on the gain of detection performance in the proposed TL2GH^{2}T. To this purpose, we validated the effectiveness of the proposed HybridHT in capturing discriminative change information by a comprehensive ablation study.

We first illustrate the effectiveness of the proposed HybridHT by examining the impact of head hybrid operations in the proposed model. Specifically, changes in detection performance (in terms of OA) are observed by setting δ to 0 to ignore the head hybrid operations. The corresponding

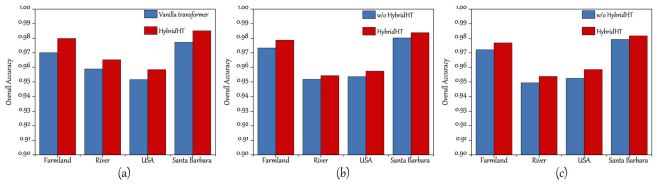


Fig. 8. Ablation analysis of the proposed HybridHT. (a) Performance of TL2GH^{2}T with or without HybridHT. (b) Performance of SSTFormer with or without HybridHT. (c) Performance of CDFormer with or without HybridHT.

experimental results on four datasets are presented in Fig. 8. As it can be observed, the head exchange operation in HybridHT effectively improves performance. It can be easily derived that the exploration of temporal correlation in the global perspective can improve detection accuracy.

In addition, we further validate the effectiveness of HybridHT by combining it with two other existing Transformer-based HSI-CD models (i.e., SSTFormer and CDFormer). To be more precise, we replace the HybridHT suggested in this study with the change feature extraction module (i.e., T-Former in the SSTFormer and vanilla transformer in the CDFormer) used in these two models and the experimental results are illustrated in Fig. 8. It is evident that applying the proposed HybridHT directly to existing methods also yields a substantial gain in detection performance. Comprehensively, the introduced HybridHT not only boosts the CD performance of the proposed TL2GH^{2}T but also effectively enhances the detection accuracy of other transformer-based models.

E. Qualitatively Evaluation

Table II records the mean value and standard deviation of OA and Kappa coefficient obtained by the aforementioned state-of-the-art methods and our proposed TL2GH^{2}T across four datasets. As far as conventional detectors are concerned, i.e., CVA and its variant PCA-CVA unsurprisingly achieve poor performance due to the fact that they simply work by comparing spectral variability, which is highly dependent on the threshold setting. Remarkable performance is appropriately accomplished using the strong feature representation powers inherent in DL technologies, such as CNN- and transformer-based techniques. Furthermore, as a burgeoning technology in contemporary years, the transformer-based method often performs better in terms of detection accuracy than the CNN-based methods, thanks to its global feature extraction capabilities. In addition, given that SST-Former independently explores the spatial–spectral features of the two temporal images using a spatial–spectral transformer through the transformer encoder before engaging in change feature extraction, it also exhibits superior detection performance compared with CDFormer.

Notably, we can observe that the proposed TL2GH^{2}T is superior to other compared methods in terms of the two above evaluation indices. Specifically, the OA outperformed the second-best CD algorithm, by 0.45%, 1.32%, 0.48%, and 0.49%, respectively, with the exception of TL2GH^{2}T. It is worth noting that none of the other state-of-the-art methods performed well enough on the River dataset. This was mostly because there were so many regions with subtle variations in the image which made accurate detection difficult. In contrast,

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TABLE II
QUANTITATIVE COMPARISON OF DIFFERENT STATE-OF-THE-ART DETECTORS IN TERMS OF OA (%) AND KAPPA COEFFICIENT ON FOUR HS DATASETS. THE BOLD ENTITIES INDICATE THE BEST ONE

Data Set	Method	CVA	PCA-CVA	Re3FCN	SSDIN	GETNET	TDSSC	SST-Former	CDFormer	TL2GH^{2}T
Farmland	OA (%)	95.85	97.53	96.37±0.22	95.03±0.16	96.84±0.19	97.23±0.27	97.33±0.18	97.21±0.22	**97.98±0.07**
	Kappa×100	89.88	93.98	91.02±0.55	87.51±0.46	91.85±0.27	93.12±0.44	93.62±0.28	93.47±0.48	**94.99±0.13**
	Total Params	/	/	72.680K	**44.936K**	91.23M	199.714K	2.498M	63.559M	174.208K
	Total Flops	/	/	12.499G	7.540G	53.79G	10.131G	9.233G	8.059G	**1.571G**
River	OA (%)	93.05	95.2	93.35±0.28	94.98±0.43	93.91±0.31	94.54±0.41	95.18±0.34	94.94±0.45	**96.52±0.24**
	Kappa×100	66.91	66.61	51±0.69	71.13±0.87	66.89±0.59	69.87±0.75	72.35±0.69	71.75±1.05	**77.12±0.67**
	Total Params	/	/	73.454K	**45.710K**	150.566M	246.670K	2.520M	81.191M	174.208K
	Total Flops	/	/	15.967G	9.631G	88.78G	14.465G	9.427G	10.295G	**1.571G**
USA	OA (%)	93.47	92.53	89.97±0.24	94.75±0.36	93.48±0.23	93.68±0.31	95.36±0.27	95.25±0.35	**95.84±0.17**
	Kappa×100	80.33	76.04	71.21±0.43	85.02±0.65	84.32±0.61	84.68±0.69	86.89±0.63	85.88±0.64	**87.91±0.58**
	Total Params	/	/	72.662K	**44.918K**	90.02M	198.622K	2.498M	63.149M	174.208K
	Total Flops	/	/	12.419G	7.491G	53.07G	10.127G	9.229G	8.007G	**1.571G**
SantaBarbara	OA (%)	83.4	89.82	97.75±0.27	97.55±0.34	97.83±0.29	97.89±0.39	98.02±0.25	97.92±0.24	**98.51±0.14**
	Kappa×100	65.79	78.63	95.56±0.77	94.89±0.73	96.12±0.54	95.26±0.61	96.8±0.56	95.65±0.67	**96.87±0.55**
	Total Params	/	/	73.922K	**46.178K**	193.623M	294.718K	2.534M	91.853M	208.288K
	Total Flops	/	/	18.064G	10.896G	114.169G	16.142G	9.543G	11.647G	**4.333G**

the proposed TL2GH^{2}T demonstrates superior performance in the challenging scenario of the River dataset due to its comprehensive exploration of spatial and spectral features from local-to-global perspectives. Another interesting phenomenon is that despite the more powerful feature extraction capabilities of DL-based methods, none of them, except the proposed TL2GH^{2}T, outperforms the machine-learning-based PCA-CVA on the Farmland and Rivers datasets. This might be carefully explained by the limited sample size provided by these two datasets, which is detrimental to DL-based techniques that depend on sample size. On the other hand, this implies that our proposed method demonstrates better feature mining efficiency with small samples.

Moreover, we conducted an additional assessment of the computational complexity by considering both trainable parameters and floating point operations (FLOPs), for various DL-based methods using the Farmland dataset. As reported in Table II, it is evident that the most of the CNN-based detection methods possess fewer parameters when compared with the transformer-based methods. Because affinity matrices constructed from spectral features are used as model inputs and have a larger spatial dimension by nature, GETNET contains more parameters. It is widely known that transformer-based approaches are well-known to introduce a large number of parameters and FLOPs, which pose significant challenges to the computational efficiency of the models. However, compared with fully used transformer-structured models (such as SST-Former and CDFormer), TL2GH^{2}T proposed in this study not only demonstrates a notable lightweight advantage but also maintains its competitiveness among CNN-based models.

F. Visual Evaluation

To visually compare the output of various methods, we implemented ten independent repetitions of the experiment on the four datasets using the proposed TL2GH^{2}T and the detectors mentioned above. The best detection graph for each algorithm was then selected based on OA and compared with the ground-truth graph of the datasets, as shown in Figs. 9–12. The detection map shown above makes it intuitively clear that when compared with other approaches used for comparison, TL2GH^{2}T described in this article exhibits a detection performance that is closer to the ground truth. Notably, one of the challenges in the HSI-CD task is to detect subtle changes in land covers while avoiding being affected by pseudochanges due to spectral variability. For instance, the raw spectral change-based detection maps produced by CVA algorithms typically show a greater proportion of overdetected land coverings. While PCA-CVA uses PCA operation to minimize the impact of spectral pseudovariations, as a result, it overlooks

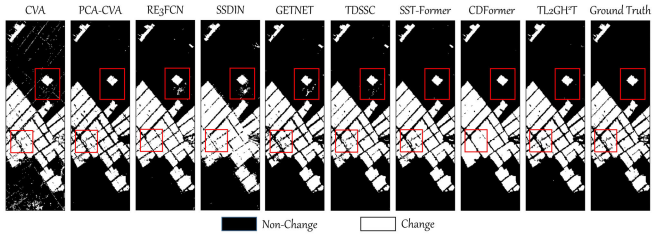


Fig. 9. CD maps generated by different methods on the Farmland datasets. The areas within red squares indicate the regions of interest.

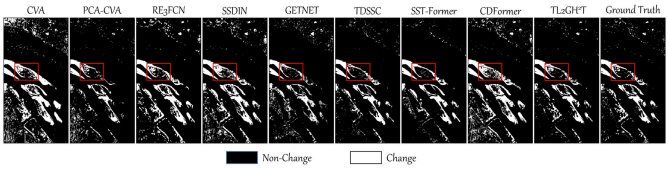


Fig. 10. CD maps generated by different methods on the River datasets. The area within the red square indicates the region of interest.

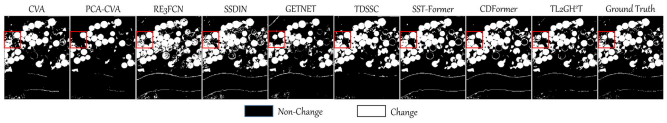


Fig. 11. CD maps generated by different methods on the USA datasets. The area within the red square indicates the region of interest.

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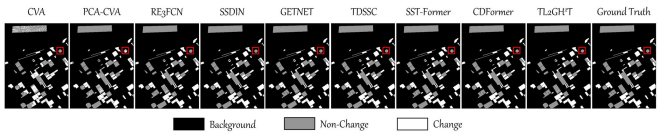


Fig. 12. CD maps generated by different methods on the Santa Barbara datasets. The area within the red square indicates the region of interest.

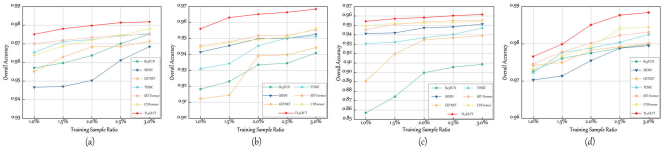


Fig. 13. Classification performance (OA) obtained by different methods with varying percentages of training samples on (a) Farmland dataset, (b) River dataset, (c) USA dataset, and (d) Santa Barbara dataset.

a significant number of subtle changes. On the basis of comparing deep spectral features, GETNET also produces salt and pepper noise in the detection maps, particularly in the River dataset. Furthermore, Re3FCN, SSDIN, TDSSC, SST-Former, and CDFormer are a few examples of methods that improve the extraction of spatial semantic information by including spatial aspects to produce visually clearer detection results. Unfortunately, the detection maps obtained by these methods generally suffer from oversmoothing and at the same time fail to detect many subtle changes. The reason for this phenomenon is primarily attributed to these methods' heavy reliance on spatial information differences as the basis for pixel-level CD, resulting in overly homogeneous change areas. As the selected regions of interest from Figs. 9 to 12 indicated, the proposed TL2GH²T achieves the most desirable detection maps, efficiently recognizing both boundary details and tiny land cover changes.

G. Impact of the Number of Training Samples

Considering the scarcity of labeled samples in practical application scenarios, we further investigated the detection performance of all the DL-based algorithms for different training sample sizes over four datasets. To be more precise, we randomly selected a diverse percentage of annotated samples from both the change and unchanged regions on every dataset in the proportion of $[1\%, 1.5\%, \ldots, 2.5\%]$ at intervals of 0.5% as the training set. Fig. 13 displays the OA achieved by different methods under the variations in sample size. As expected, the increase in training sample size is accompanied by a rise in detection accuracy. However, as the sample becomes more homogeneous within the class, the rate of increase usually decreases. Remarkably, the proposed TL2GH²T demonstrates superior performance over all the comparative methods in detection, regardless of the number of labeled samples provided for model training.

V. Conclusion

In this article, we present a novel HSI-CD framework, i.e., TL2GH²T, by thoroughly considering the diversity of diagnostic features in HS images and the significance of deep global temporal information. To simultaneously explore spatial, spectral, and spatial–spectral features, we introduce TL2GN, aiming to generate discriminative features from bitemporal HS images. Furthermore, we customized a fusion module, called SSTF, to adaptively fuse these obtained features for further facilitated extraction of temporal correlations. Moreover, inspired by the concept of chromosomal crossover, the proposed HybridHT is capable enough of capturing the abstract temporal properties of bitemporal images effectively for the CD task, making it more applicable to various real-world scenarios. Extensive experiments conducted on four real datasets demonstrate the superiority of TL2GH²T over current state-of-the-art methods in HSI-CD tasks.

In the next phase of our work, methods for extracting temporal difference features deserve further investigation. In addition to extracting feature-level correlations, we also intend to consider correlations between SA weights for different temporal images in future studies.

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